

Operations Research

Problem Set 3 – Detailed Solutions

Notation as in Lectures 2 and 4. For $B \subseteq \{1, \dots, n\}$, A_B is the submatrix of the columns indexed by B ; a basis is a B with $|B| = m$ and A_B invertible; the basic solution of B has $x_N = 0$ and $x_B = A_B^{-1}b$, and is a BFS if $x_B \geq 0$. The support of x is $\text{supp}(x) = \{j : x_j \neq 0\}$.

Problem 1 – An optimal basic feasible solution exists

Problem 1

$A \in \mathbb{R}^{m \times n}$ has rank m , $Q = \{x : Ax = b, x \geq 0\} \neq \emptyset$, and $\sup_{x \in Q} c^\top x < +\infty$. Prove that some BFS $x^* \in Q$ satisfies $c^\top x^* \geq c^\top x$ for every $x \in Q$, without assuming the supremum is attained.

Solution 1

The proof has two steps. First, every feasible point can be replaced by a BFS that is at least as good. Second, there are only finitely many BFSs, so the best of them is optimal.

Step 1: from a feasible point to a BFS of no smaller value.

Claim. For every $x \in Q$ there is a BFS $x' \in Q$ with $c^\top x' \geq c^\top x$.

Proof of the claim. We argue by induction on $|\text{supp}(x)|$. Let $x \in Q$, $S = \text{supp}(x)$, so $x_j > 0$ for $j \in S$ and $x_j = 0$ otherwise.

Case 1: the columns A_S are linearly independent. Since $\text{rank } A = m$, the independent set S can be extended to a set $B \supseteq S$ of m independent columns; B is a basis. The vector x has $x_j = 0$ for $j \notin B$ and $A_B x_B = Ax = b$. As A_B is invertible, $x_B = A_B^{-1}b$ is the basic solution of B , and it is nonnegative. Hence x itself is a BFS; take $x' = x$. (If $S = \emptyset$, i.e. $x = 0$, the empty set is independent and the same argument applies.)

Case 2: the columns A_S are linearly dependent. Then there is a nonzero $d \in \mathbb{R}^n$ with $\text{supp}(d) \subseteq S$ and $Ad = 0$. Replacing d by $-d$ if necessary we may assume $c^\top d \geq 0$. For every $\theta \geq 0$ put

$$x(\theta) = x + \theta d.$$

Then $Ax(\theta) = Ax + \theta Ad = b$, $x(\theta)_j = 0$ for $j \notin S$, and $c^\top x(\theta) = c^\top x + \theta c^\top d \geq c^\top x$. We distinguish two sub-cases.

Case 2(i): d has a negative component. Let

$$\theta^* = \min_{j \in S: d_j < 0} \frac{x_j}{-d_j} > 0.$$

For j with $d_j < 0$, $x(\theta^*)_j = x_j + \theta^* d_j \geq x_j - \frac{x_j}{-d_j}(-d_j) = 0$, with equality for a minimising index; for j with $d_j \geq 0$, $x(\theta^*)_j \geq x_j \geq 0$. So $x(\theta^*) \in Q$, $c^\top x(\theta^*) \geq c^\top x$, and at least one index of S has become zero while no new nonzero index appeared: $|\text{supp}(x(\theta^*))| < |\text{supp}(x)|$. By the induction hypothesis there is a BFS x' with $c^\top x' \geq c^\top x(\theta^*) \geq c^\top x$.

Case 2(ii): $d \geq 0$ (and $d \neq 0$). Then $x(\theta) \geq 0$ for every $\theta \geq 0$, so $x(\theta) \in Q$ for all $\theta \geq 0$. If $c^\top d > 0$, then $c^\top x(\theta) = c^\top x + \theta c^\top d \rightarrow +\infty$, contradicting $\sup_Q c^\top x < +\infty$. Hence $c^\top d = 0$. Now use $-d$ instead: $-d \neq 0$ has a negative component, $A(-d) = 0$, $\text{supp}(-d) \subseteq S$, and $c^\top(-d) = 0 \geq 0$. Case 2(i) applied to $-d$ gives a point of Q with strictly smaller support and the same objective value, and the induction hypothesis finishes as before.

The induction is on $|\text{supp}(x)| \in \{0, 1, \dots, n\}$; the base case $|\text{supp}(x)| = 0$ falls under Case 1. This proves the claim. $\square$

Step 2: the best BFS is optimal. There are at most $\binom{n}{m}$ bases, and each basis has exactly one basic solution, so the set $\mathcal{F}$ of BFSs is finite. It is nonempty: $Q \neq \emptyset$ and the claim applied

to any $x \in Q$ produces a BFS. A finite nonempty set of real numbers has a maximum, so we may choose

$$x^* \in \mathcal{F} \quad \text{with} \quad c^\top x^* = \max_{x' \in \mathcal{F}} c^\top x'.$$

Let $x \in Q$ be arbitrary. By the claim there is $x' \in \mathcal{F}$ with $c^\top x' \geq c^\top x$, and by the choice of x^* , $c^\top x^* \geq c^\top x'$. Therefore $c^\top x^* \geq c^\top x$ for every $x \in Q$.

In particular the supremum is attained: $\sup_Q c^\top x = c^\top x^* = \max_Q c^\top x$. We never assumed attainment; it followed from the finiteness of $\mathcal{F}$. ■

Remark

The proof is Problem 6(c) of Set 1 (“ Q contains a BFS”) with one extra ingredient: the direction d is chosen so that moving along it does not decrease $c^\top x$, and the bounded-supremum hypothesis rules out the one situation, $d \geq 0$ with $c^\top d > 0$, where the move would never stop. Without that hypothesis the conclusion is false: $\max x_1$ s.t. $x_1 - x_2 = 0$, $x \geq 0$ has $Q \neq \emptyset$, a unique BFS $(0, 0)$, and no maximum.

Problem 2 – The dual, and the dual of the dual

Problem 2

(P): $\max c^\top x$ s.t. $Ax \leq b$, $x \geq 0$. (a) Derive the dual (D). (b) Prove that the dual of (D) is (P).

Solution 2(a)

We look for the best *upper bound* on $c^\top x$ that can be proved by combining the constraints of (P) linearly, as in Lecture 4.

Constraints and variable domains. Choose a weight y_i for each of the m inequalities $a_i^\top x \leq b_i$ (where $a_i^\top$ is row i of A). Multiplying an inequality by a weight preserves its direction exactly when the weight is nonnegative; so we require $y_i \geq 0$ for $i = 1, \dots, m$. Multiplying and adding,

$$\sum_{i=1}^m y_i a_i^\top x \leq \sum_{i=1}^m y_i b_i, \quad \text{i.e.} \quad (A^\top y)^\top x \leq b^\top y \quad \text{for every feasible } x.$$

The left side is $\sum_{j=1}^n (A^\top y)_j x_j$. We want it to dominate $c^\top x = \sum_j c_j x_j$. Since every feasible x has $x_j \geq 0$, it suffices that each coefficient dominates: $(A^\top y)_j \geq c_j$ for $j = 1, \dots, n$, that is $A^\top y \geq c$. Under these conditions,

$$c^\top x \leq (A^\top y)^\top x \leq b^\top y \quad \text{for every feasible } x.$$

Objective. Every $y \geq 0$ with $A^\top y \geq c$ certifies the upper bound $b^\top y$ on the optimal value of (P). The best such certificate is the smallest bound. Hence the dual is

$$(D) \quad \min b^\top y \quad \text{s.t.} \quad A^\top y \geq c, \quad y \geq 0.$$

It has m variables (one per primal constraint) and n constraints (one per primal variable), and its constraint matrix is $A^\top$ because the coefficient of x_j in the combined inequality is the j -th column of A dotted with y .

Solution 2(b)

The construction of (a) applies to a maximisation with $\leq$ constraints and nonnegative variables. We rewrite (D) in that shape, dualise, and rewrite back. Each rewriting is an equivalence of optimisation problems: same feasible set, and objectives that differ only by a sign, so the same optimal solutions.

Step 1: (D) as a maximisation of the required shape. Minimising $b^\top y$ is the same as maximising $-b^\top y$, and $A^\top y \geq c$ is the same as $(-A^\top)y \leq -c$. So (D) is equivalent to

$$(D') \quad \max (-b)^\top y \quad \text{s.t.} \quad (-A^\top)y \leq -c, \quad y \geq 0,$$

which has the form of (P) with the data (A, b, c) replaced by $(-A^\top, -c, -b)$: the matrix is $-A^\top \in \mathbb{R}^{n \times m}$, the right-hand side is $-c \in \mathbb{R}^n$, the objective vector is $-b \in \mathbb{R}^m$.

Step 2: apply (a) to (D'). By (a), the dual of a program $\max \tilde{c}^\top y$ s.t. $\tilde{A}y \leq \tilde{b}, y \geq 0$ is $\min \tilde{b}^\top x$ s.t. $\tilde{A}^\top x \geq \tilde{c}, x \geq 0$, with one variable x_j per constraint of (D'), that is $x \in \mathbb{R}^n$. With $\tilde{A} = -A^\top, \tilde{b} = -c, \tilde{c} = -b$ this reads

$$\min (-c)^\top x \quad \text{s.t.} \quad (-A^\top)^\top x \geq -b, \quad x \geq 0,$$

that is,

$$\min -c^\top x \quad \text{s.t.} \quad -Ax \geq -b, \quad x \geq 0.$$

Step 3: rewrite back. Minimising $-c^\top x$ is maximising $c^\top x$, and $-Ax \geq -b$ is $Ax \leq b$. The program obtained is

$$\max c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0,$$

which is (P). Hence the dual of the dual is the primal. ■

The derivation also shows the correspondence of the sign rules: a $\geq$ constraint of a minimisation becomes, after the sign flip, a $\leq$ constraint of a maximisation, and therefore produces a nonnegative dual variable; this is exactly why the variables of the dual of (D) are the nonnegative x_j of (P).

Problem 3 – Weak duality and its consequences**Problem 3**

For the pair (P), (D): (a) $c^\top x \leq b^\top y$ for all feasible x, y . (b) If feasible x, y satisfy $c^\top x = b^\top y$, both are optimal. (c) If (P) is unbounded above then (D) is infeasible; if (D) is unbounded below then (P) is infeasible.

Solution 3(a)

Let x be primal feasible ($Ax \leq b, x \geq 0$) and y dual feasible ($A^\top y \geq c, y \geq 0$). We compare both objectives with the number $y^\top Ax$.

First inequality. $A^\top y - c \geq 0$ componentwise and $x \geq 0$, so the inner product of these two nonnegative vectors is nonnegative:

$$(A^\top y - c)^\top x \geq 0 \iff c^\top x \leq (A^\top y)^\top x = y^\top Ax.$$

Second inequality. $b - Ax \geq 0$ componentwise and $y \geq 0$, so

$$y^\top (b - Ax) \geq 0 \iff y^\top Ax \leq y^\top b = b^\top y.$$

Chaining, $c^\top x \leq y^\top Ax \leq b^\top y$. ■

Solution 3(b)

Let x^*, y^* be feasible with $c^\top x^* = b^\top y^*$.

x^* is optimal for (P). For every primal feasible x , part (a) applied to the pair (x, y^*) gives $c^\top x \leq b^\top y^* = c^\top x^*$. So no feasible x beats x^* .

y^* is optimal for (D). For every dual feasible y , part (a) applied to the pair (x^*, y) gives $b^\top y \geq c^\top x^* = b^\top y^*$. So no feasible y beats y^* . ■

In words: y^* is a certificate that nothing above $c^\top x^*$ is achievable, and x^* is a certificate that nothing below $b^\top y^*$ is achievable; when the two numbers coincide, each solution proves the other optimal.

Solution 3(c)

(P) unbounded $\Rightarrow$ (D) infeasible. Suppose, for contradiction, that (D) has a feasible point $\bar{y}$. By (a), $c^\top x \leq b^\top \bar{y}$ for every primal feasible x : the primal objective is bounded above by the fixed number $b^\top \bar{y}$. This contradicts the assumption that (P) is unbounded above (that is, that feasible x with arbitrarily large $c^\top x$ exist). Hence (D) has no feasible point.

(D) unbounded below $\Rightarrow$ (P) infeasible. Symmetrically, suppose (P) has a feasible point $\bar{x}$. By (a), $b^\top y \geq c^\top \bar{x}$ for every dual feasible y , so the dual objective is bounded below by $c^\top \bar{x}$, contradicting unboundedness below. Hence (P) has no feasible point. ■

Note the converses fail: an infeasible dual does not force an unbounded primal, since both programs may be infeasible (Lecture 4 gives $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq -1$, $-x_1 + x_2 \leq -1$, $x \geq 0$, whose dual is also infeasible).

Problem 4 – Strong duality**Problem 4**

Assume (P) is feasible and $c^\top x$ is bounded above on its feasible set. Prove that both (P) and (D) attain their optima and that $\max_{Ax \leq b, x \geq 0} c^\top x = \min_{A^\top y \geq c, y \geq 0} b^\top y$. In particular, show that some dual feasible y attains the primal optimal value.

Solution 4 – overview

Weak duality (Problem 3) gives “ $\leq$ ”. The work is to produce a dual feasible y whose value equals the primal optimum. We proceed in three steps.

1. **Primal attainment.** Problem 1 gives an optimal solution x^* of (P); write $\gamma = c^\top x^*$.
2. **A theorem of the alternative (Farkas).** A system of linear inequalities has no solution exactly when a nonnegative combination of its rows reads “ $0 \leq$ negative number”. We prove this from Fourier–Motzkin elimination, which is nothing but the “combine inequalities with nonnegative weights” of Lecture 3 applied systematically.
3. **Dual attainment.** Suppose no dual feasible y has $b^\top y \leq \gamma$. Farkas turns that supposition into a vector which, read in primal terms, is either a feasible point better than x^* or a direction of unbounded improvement; both contradict the hypotheses.

Solution 4 – Step 1: the primal attains its maximum

Introduce slacks: (P) is equivalent to

$$\max c^\top x \quad \text{s.t.} \quad Ax + s = b, \quad x \geq 0, s \geq 0,$$

in the sense that x is feasible for (P) iff $(x, b - Ax)$ is feasible for the equational program, with the same objective. The constraint matrix $[A \ I_m] \in \mathbb{R}^{m \times (n+m)}$ contains I_m , so it has rank m . The equational program is feasible (because (P) is) and its objective is bounded above (because (P) 's is). By Problem 1, it has an optimal solution (x^*, s^*) , which is moreover a BFS. Then x^* is feasible for (P) and, for every feasible x of (P) , $c^\top x = c^\top(x, b - Ax) \leq c^\top x^*$. So (P) attains its maximum; set

$$\gamma = c^\top x^* = \max\{c^\top x : Ax \leq b, x \geq 0\}.$$

Solution 4 – Step 2: Fourier–Motzkin elimination and the Farkas lemma

Lemma 1 (Fourier–Motzkin elimination). *Let $M \in \mathbb{R}^{p \times k}$ with $k \geq 1$ and $d \in \mathbb{R}^p$. There exist a matrix $M' \in \mathbb{R}^{p' \times (k-1)}$ and $d' \in \mathbb{R}^{p'}$ such that*

- (i) *every row of the system $M'x' \leq d'$ is a nonnegative combination of rows of $Mx \leq d$ in which the variable x_1 has been eliminated: for each row ℓ of M' there is $\lambda^{(\ell)} \geq 0$ in $\mathbb{R}^p$ with $\lambda^{(\ell)\top} M = (0, M'_{\ell,\cdot})$ and $\lambda^{(\ell)\top} d = d'_\ell$;*
- (ii) *$Mx \leq d$ has a solution $x = (x_1, x') \in \mathbb{R}^k$ if and only if $M'x' \leq d'$ has a solution $x' \in \mathbb{R}^{k-1}$.*

Proof. Write row i of $Mx \leq d$ as $m_{i1}x_1 + m_i^\top x' \leq d_i$, where $m_i \in \mathbb{R}^{k-1}$ collects the remaining coefficients. Partition the rows into $I_+ = \{i : m_{i1} > 0\}$, $I_- = \{i : m_{i1} < 0\}$, $I_0 = \{i : m_{i1} = 0\}$. Dividing by $|m_{i1}|$, a row of I_+ says $x_1 \leq U_i(x') := (d_i - m_i^\top x')/m_{i1}$ and a row of I_- says $x_1 \geq L_i(x') := (d_i - m_i^\top x')/(-m_{i1})$. Define the new system to consist of

- the rows of I_0 unchanged (they do not involve x_1), and
- for each pair $(i, h) \in I_- \times I_+$, the inequality $L_i(x') \leq U_h(x')$, i.e.

$$\frac{1}{-m_{i1}}(m_i^\top x' - d_i) + \frac{1}{m_{h1}}(m_h^\top x' - d_h) \leq 0.$$

(i) A row of I_0 is the combination with $\lambda = e_i$. The pair row is $\frac{1}{-m_{i1}}(\text{row } i) + \frac{1}{m_{h1}}(\text{row } h)$, both weights positive, and the coefficient of x_1 is $\frac{m_{i1}}{-m_{i1}} + \frac{m_{h1}}{m_{h1}} = -1 + 1 = 0$. So every new row is a nonnegative combination of old rows with x_1 eliminated.

(ii), “only if”. If (x_1, x') solves the original system, the I_0 rows hold for x' , and for each pair $L_i(x') \leq x_1 \leq U_h(x')$, so the pair rows hold too.

(ii), “if”. Let x' solve the new system. Put $L = \max_{i \in I_-} L_i(x')$ (or $-\infty$ if $I_- = \emptyset$) and $U = \min_{h \in I_+} U_h(x')$ (or $+\infty$ if $I_+ = \emptyset$). The pair rows say exactly $L \leq U$ when both index sets are nonempty, and $L \leq U$ holds trivially otherwise. Choose any real $x_1 \in [L, U]$. Then every row of I_- holds ($x_1 \geq L \geq L_i(x')$), every row of I_+ holds ($x_1 \leq U \leq U_h(x')$), and the I_0 rows hold by assumption. So (x_1, x') solves the original system. $\square$

Theorem 1 (Farkas lemma, inequality form). *Let $M \in \mathbb{R}^{p \times k}$ and $d \in \mathbb{R}^p$. Exactly one of the following holds:*

- (A) *the system $Mx \leq d$ has a solution $x \in \mathbb{R}^k$;*
- (B) *there is $\lambda \in \mathbb{R}^p$ with $\lambda \geq 0$, $\lambda^\top M = 0$ and $\lambda^\top d < 0$.*

Proof. Not both. If x solves (A) and λ satisfies (B), then $0 = (\lambda^\top M)x = \lambda^\top(Mx) \leq \lambda^\top d < 0$, where the inequality uses $Mx \leq d$ and $\lambda \geq 0$. Contradiction.

At least one. Suppose (A) fails. Apply Lemma 1 k times, eliminating $x_1, x_2, \dots, x_k$ in turn. By (ii), unsolvability is preserved at every step, so the final system, which has no variables, is unsolvable. A system with no variables consists of inequalities of the form $0 \leq d''_\ell$; it is

unsolvable exactly when some $d''_\ell < 0$. By (i), applied at each step and composed (a nonnegative combination of nonnegative combinations is a nonnegative combination), that row is $\lambda^\top (Mx \leq d)$ for some $\lambda \geq 0$ with $\lambda^\top M = 0$ and $\lambda^\top d = d''_\ell < 0$. This is (B). $\square$

Solution 4 – Step 3: the dual attains the primal value

We show that the system

$$(*) \quad A^\top y \geq c, \quad y \geq 0, \quad b^\top y \leq \gamma$$

has a solution. Written in the form $My \leq d$ of Theorem 1, with $y \in \mathbb{R}^m$,

$$M = \begin{pmatrix} -A^\top \\ -I_m \\ b^\top \end{pmatrix} \in \mathbb{R}^{(n+m+1) \times m}, \quad d = \begin{pmatrix} -c \\ 0 \\ \gamma \end{pmatrix}.$$

Suppose $(*)$ has no solution. By Farkas there is $\lambda = (u, w, t) \geq 0$ with $u \in \mathbb{R}^n$, $w \in \mathbb{R}^m$, $t \in \mathbb{R}$ such that $\lambda^\top M = 0$ and $\lambda^\top d < 0$:

$$-Au - w + tb = 0 \quad \text{and} \quad -c^\top u + t\gamma < 0, \quad \text{i.e.} \quad Au + w = tb \quad \text{and} \quad c^\top u > t\gamma.$$

Since $w \geq 0$, the first relation gives $Au \leq tb$. Two cases.

Case $t > 0$. Put $x = u/t \geq 0$. Then $Ax \leq b$, so x is feasible for (P) , and $c^\top x = c^\top u/t > \gamma$. This contradicts $\gamma = \max_{(P)} c^\top x$ (Step 1).

Case $t = 0$. Then $Au \leq 0$, $u \geq 0$ and $c^\top u > 0$. For every $\theta \geq 0$ the point $x^* + \theta u$ is feasible for (P) : it is nonnegative, and $A(x^* + \theta u) = Ax^* + \theta Au \leq b$. Its objective is $c^\top x^* + \theta c^\top u \rightarrow +\infty$ as $\theta \rightarrow \infty$, contradicting the boundedness of $c^\top x$ on the feasible set.

Both cases are impossible, so $(*)$ has a solution y^* : it is dual feasible and $b^\top y^* \leq \gamma$. By weak duality (Problem 3(a)) with the primal optimum x^* , $b^\top y^* \geq c^\top x^* = \gamma$. Hence

$$b^\top y^* = \gamma = c^\top x^*.$$

Solution 4 – conclusion

We have a primal feasible x^* and a dual feasible y^* with equal objective values. By Problem 3(b) both are optimal. Therefore (P) attains its maximum at x^* , (D) attains its minimum at y^* , and

$$\max_{Ax \leq b, x \geq 0} c^\top x = c^\top x^* = b^\top y^* = \min_{A^\top y \geq c, y \geq 0} b^\top y.$$

The dual feasible solution y^* of Step 3 is the required certificate attaining the primal optimal value. $\blacksquare$

Remark

Where each hypothesis was used. Feasibility of (P) and boundedness of $c^\top x$ entered in Step 1 (to invoke Problem 1) and again in Step 3 (to refute both Farkas cases). The rank condition of Problem 1 was supplied for free by the slack columns. Nothing was assumed about (D) : its feasibility and the attainment of its minimum were *derived*.

Why Farkas is the right tool. Weak duality says that any true upper bound of the form “ $c^\top x \leq \beta$ ” that follows from nonnegative combinations of the constraints is certified by a dual feasible y . Strong duality says the *best* bound, γ , is also of that form. Farkas is precisely the statement that for linear inequalities, whatever is true is provable by nonnegative combination, and Fourier–Motzkin is the procedure that finds the combination.